

SoftWall: Certifying Conditional Recovery for Shared-GPU AI-RAN

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AI-and-RAN platforms seek to run radio processing and external AI services on shared accelerators. GPU concurrency mechanisms such as NVIDIA Multi-Process Service (MPS), however, do not by themselves provide a temporal contract for the radio path. The problem becomes sharper when a neural receiver is optional: until its result is known, the same transport block retains a conventional-receiver recovery obligation. Across cells and RAN homes, these contingent obligations can collide on shared recovery resources even though they have not entered a runtime queue.

We present SoftWall, a runtime substrate that represents unresolved conventional work as recovery debt. SoftWall continuously maintains an executable all-fail schedule for every unresolved neural receiver. It may retime recovery intervals and admit a bounded external-AI unit only through one generation-safe transaction that preserves this certificate. The transaction covers endpoint and transport credits, external-AI ownership, physical latest-start checks, CUDA completion fences, and single radio commit. We also derive a provenance-qualified feasibility envelope that distinguishes safe-and-useful, safe-without-slack, mandatory-infeasible, and unqualified modes.

Two prespecified boundary states show why current GPU idleness is insufficient: debt-blind AI admission can cross the radio guard by 12 ms and 1 ms under the qualified contract, while SoftWall rejected all 60 corresponding requests before physical launch. On two independent 4×A100 nodes with MIG disabled and MPS enabled, a warm path executed 4,800 actual TensorRT neural-receiver requests, 1,312 shared-cuPHY recoveries, 1,077 Qwen units, and 4,800 radio commits without an observed deadline, bound, transport, credit, or commit violation. A seven-class fault campaign executed another 2,800 neural-receiver requests and 942 recoveries with no observed deadline miss. The model agreed with an exact checker on 16,023 qualified states and with 180 prespecified physical boundary rounds; a certified 64-debt scheduler had 1.494 ms p99 decision latency. A strong certificate-preserving recovery-first baseline and an offline oracle found no material throughput advantage, so we make no optimizer claim. SoftWall instead identifies when conditional sharing is certifiably safe, when it has no useful slack, and when the mode must be rejected or requalified. These timing claims are finite-sample qualifications of synthetic P180/D155 modes. Applying the same discipline to Aerial testMAC's 4.5 ms UL-indication threshold rejected every tested implementation; persistent input assembly and stream ordering improved the best raw-IQ two-GPU prototype to 995/1,000 clean requests, but the frozen development gate still failed and did not open a holdout. Four of the five late samples were limited by the conventional path and one by remote NeuralRx. External Aerial TDL-A remains unqualified, but a prespecified disjoint-seed Sionna CDL-D/E holdout passed: among 500 paired trials, low-SNR outcomes contained 31 NeuralRx-only versus 12 conventional-only correct blocks. These gates bound the present claim and identify the required production path and supported channel domain.

CCS Concepts: • **Computer systems organization** → **Distributed architectures; Real-time systems; Networks** → *Network resources allocation*.

Additional Key Words and Phrases: AI-RAN, GPU scheduling, neural receiver, conditional recovery, MPS, feasibility certificate

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1 Introduction

Modern base stations increasingly use GPUs for both the physical layer and AI services. This creates an appealing utilization opportunity: serve inference requests while radio demand leaves compute idle. Existing AI-and-RAN systems allocate GPU fractions, place work across accelerator pools,

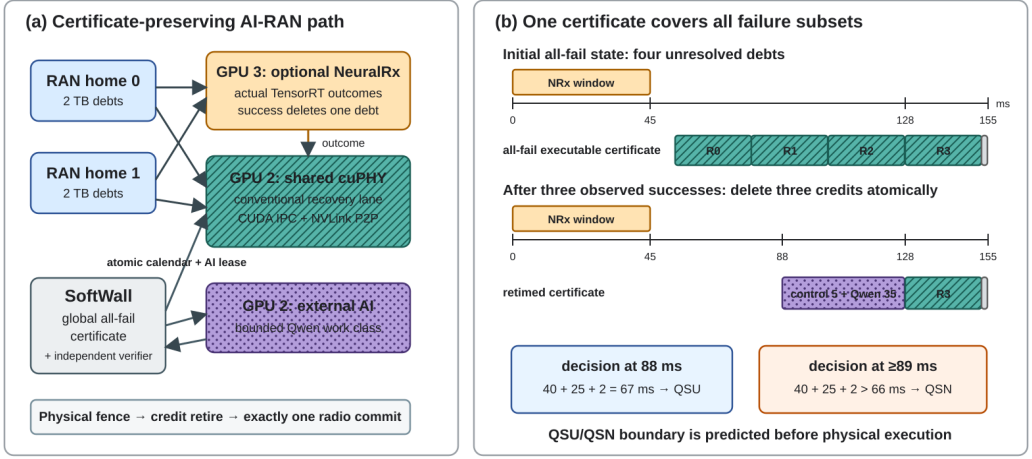


Fig. 1. SoftWall maintains one global all-fail certificate across two RAN homes, optional NeuralRx, shared cuPHY recovery, and bounded external AI. Three observed successes delete three recovery credits atomically. Under the illustrated qualified bounds, a decision at 88 ms admits the transaction while the adjacent 89 ms state rejects it before GPU launch.

reclaim slack, or split background work into bounded units [1, 10, 11, 13]. Those mechanisms are valuable, but they do not resolve a semantic dependency introduced by an optional neural receiver.

Consider an uplink transport block (TB) for which a conventional receiver is mandatory and a NeuralRx path is optional. A timely successful NeuralRx result can improve radio utility. A failed, late, or lost result must still leave enough time to execute conventional decoding and publish one radio result. Before the NeuralRx outcome is known, conventional work is therefore a *future mandatory job*. It may not appear in any GPU queue, yet scheduling an external-AI kernel into its eventual recovery interval can make the radio deadline infeasible.

This is a multi-request problem. Several cells may select NeuralRx at once and fail together. Several RAN homes may send their conventional recoveries to one shared cuPHY worker. Local schedules can each look feasible while their union overloads the shared lane. For example, with 25 ms recoveries and a 100 ms deadline, homes with three and two obligations are locally feasible but require 125 ms together. GPU allocation percentages and queue-aware placement cannot detect this state unless the runtime represents obligations that have not yet materialized as queued work.

SoftWall treats each unresolved optional path as *recovery debt*. At every state transition, it retains an executable schedule for the branch in which all unresolved NeuralRx requests fail. This single all-fail certificate covers every failure subset when successful outcomes only delete mandatory work. An external-AI unit is admitted only if its complete transaction—control, physical execution, and guard—can be inserted while leaving the recovery certificate valid.

The certificate must survive implementation details. We found counterexamples in which a safe plan became stale before dispatch, simultaneous outcomes were replayed through a state that never existed, launch control was omitted from an AI lease, and a second prepare created an untracked broker-held token. SoftWall closes these gaps with bounded revalidation, atomic common-cutoff outcome batches, physical latest-start enforcement, single-token ownership, and fence-confirmed retirement. These mechanisms are conventional in isolation. Their role here is to refine one radio recovery certificate through CUDA, IPC/P2P, and distributed control state.

We make four contributions:

- (1) **Certificate-to-execution refinement.** We derive seven requirements from measured counterexamples and implement atomic recovery retiming and external-AI leasing across RAN homes, shared cuPHY recovery, local/remote NeuralRx endpoints, and physical completion lifecycles.
- (2) **Predictive feasibility envelope.** We classify qualified-safe-useful (QSU), qualified-safe-no-slack (QSN), mandatory-infeasible (MI), and unqualified (UQ) states using debt, decision time, AI class, capacity, and provenance. An independent verifier prevents heuristic schedules from causing unsafe admission.
- (3) **Conditional-recovery model.** We specialize established primary/backup reasoning to same-TB optional reception, shared recovery, and a third external-AI class. All admissible NeuralRx failure subsets reduce to one executable all-fail certificate under explicit assumptions.
- (4) **Physical evaluation and negative policy result.** We evaluate actual TensorRT NeuralRx, NVIDIA Aerial/cuPHY recovery, Qwen inference, MPS, CUDA IPC, and NVLink P2P on independent A100 nodes. Strong baselines and an offline oracle show no material throughput headroom in the evaluated trace; we therefore separate the substrate contribution from optimizer claims.

Claim boundary. SoftWall does not claim that MPS provides hard isolation, that measured maxima are worst-case execution times (WCETs), or that the synthetic deadline is a production HARQ contract. Its guarantees are conditional on an explicitly qualified mode and service bounds. Five lifecycle subsets are qualified or partially qualified for the paper; five remain UQ. The claim does not require or imply a 10/10 lifecycle matrix.

Artifact plan. Upon acceptance, we will release the controller and verifier, finite-state and exact model checkers, machine-readable protocols and analyzers, sanitized timing records, and manifests that bind source to reported results through a long-lived archival repository. Proprietary Aerial/cuPHY, TensorRT, or model components will be accompanied by acquisition and build instructions rather than redistributed.

2 Background and Problem

2.1 AI-and-RAN on shared GPUs

MPS allows CUDA processes to execute concurrently and can limit a client's active-thread percentage [7]. It does not bound host scheduling, queueing, context lifecycle, memory allocation, or arbitrary kernel blocking. In our early experiments, separate-client overrun produced deadline misses under both 100% and 20% active-thread caps. A CPU socket sham produced 0 misses in 10,000 requests, while a GPU/MPS client-lifecycle arm produced 12. We use MPS as a concurrency mechanism and construct the timing contract above it.

MIG can provide stronger spatial separation, but it fixes resource partitions and is not the object of this work. We intentionally study MIG-off modes in which conditional capacity may be reclaimed. A mode that cannot obtain qualified bounds remains UQ instead of inheriting a bound from another lifecycle or placement.

2.2 Optional NeuralRx is speculative radio execution

For one radio request, NeuralRx is not interchangeable with a background AI job. It may improve the radio result, but a failed or late NeuralRx leaves conventional decoding mandatory. A correct runtime must decide three linked questions:

- (1) Which optional NeuralRx endpoints can return before the fallback cutoff?

Table 1. Counterexamples define the required refinement chain.

Counterexample	Broken assumption	Required refinement
Local-safe/global-unsafe	Local certificates compose on a shared lane	Global obligation set and certificate
23.530 ms dispatch tail	Certificate stays fresh until launch	Revalidation and physical latest-start
Sequential cutoff replay	Simultaneous outcomes may apply in any order	Atomic outcome batch
Fixed AI lease	Execution time covers launch control	Control + AI blackout
Reply loss after effect	Retrying ambiguity is harmless	Generation quarantine and fail-closed admission
Two outstanding prepares	RPC at-most-once implies visible ownership	Single unlaunched-token invariant
Warm-bound reuse	Lifecycle does not change service tails	Provenance-scoped qualification

- (2) Can every unresolved request still recover if all accepted NeuralRx paths fail together?
 (3) Which bounded external-AI units can execute without invalidating that recovery plan?

Current queue contents answer none completely because the recovery jobs are contingent. Static worst-case reservation is safe but can retain recovery capacity after successful NeuralRx outcomes. SoftWall reclaims that capacity only after observing the corresponding physical outcome.

2.3 Threat model and qualification

Each mode specifies the GPU topology, MPS settings, process lifecycle, worker epochs, memory residency, co-run class, and finite-sample whole-path bounds. SoftWall handles NeuralRx failure and lateness, stale and duplicate messages, reply loss at bounded control points, and loss of an AI reply after physical completion. It fails closed when completion evidence is missing.

We exclude arbitrary GPU hangs, driver reset, unbounded host suspension, production-clock uncertainty, and workloads outside qualified classes. Mandatory offered load that has no all-fail schedule is reported as MI rather than rescued by dropping a safety check.

3 Failure-Derived Design Requirements

SoftWall uses known primitives: MPS, bounded admission, list scheduling, generation checks, CUDA fences, IPC/P2P rings, and at-most-once ownership. Novelty does not follow from listing them. It follows from the cross-layer invariant they refine and the measured counterexamples that break naive composition.

The refinement chain is the systems contribution: a mathematically valid radio certificate remains valid through physical GPU execution and distributed control state. The model is useful because each symbol is bound to an executable object and every admission returns a verifier-checkable witness.

4 Model

4.1 Jobs, modes, and provenance

A radio request is

$$J_i = (r_i, d_i, h_i, c_i, K_i, q_i), \quad (1)$$

where r_i is release, d_i is radio-result expiry, h_i is its recovery home, c_i is a qualified conventional host-to-commit bound, K_i is the set of eligible NeuralRx endpoints, and q_i is optional radio value. Endpoint eligibility includes queue, ring, transport, NeuralRx, observation, and commit costs under the current co-run mode.

An external-AI request is

$$A_a = (u_a, v_a, \ell_a, b_a, g_a), \quad (2)$$

where u_a is visible arrival, v_a is timely value, ℓ_a is deadline, b_a is a qualified physical-completion bound for its work class, and g_a is the guard before recovery.

A bound is usable only when its provenance tuple matches the active mode:

$$\pi = (\text{GPU}, \text{placement}, \text{MPS}, \text{lifecycle}, \text{co-run}, \text{software}). \quad (3)$$

This rule prevents a warm, garbage-collection-off measurement from silently qualifying a cold, GC-on, remote, or restarted mode.

4.2 Recovery debt and all-fail certificates

While outcome i is unresolved, it induces recovery job

$$R_i = (a_i(t), d_i - g_i, c_i, h_i). \quad (4)$$

For unresolved set $U(t)$, an all-fail certificate assigns every R_i an interval $[s_i, s_i + c_i]$ satisfying release, deadline, resource-capacity, and non-overlap constraints. Shared AI blackouts and transport credits are included wherever they contend with recovery.

PROPOSITION 1 (ALL-FAIL DOMINANCE). *If a successful NeuralRx only deletes its conventional recovery and does not increase any remaining job's demand, a feasible certificate for $U(t)$ is feasible for every failure subset $F \subseteq U(t)$.*

PROOF. Delete from the all-fail schedule the intervals corresponding to successful requests. Releases, deadlines, capacities, and non-overlap of the retained intervals are unchanged. $\square$

The reduction avoids enumerating $2^{|U|}$ failure outcomes. It does not apply when success creates new mandatory work or changes another recovery's bound; such a mode needs a different proof.

4.3 Shared recovery and atomic state

Certificates over disjoint recovery resources compose by union. They do not compose when homes share a recovery GPU. SoftWall therefore constructs a global certificate over the union of obligations and commits all obligation, calendar, and AI-lease changes in one generation. The three-home plus two-home example in Section 1 is a minimal local-safe/global-unsafe counterexample.

The runtime state is

$$X_t = (S_t, Q_t, T_t, L_t, E_t, C_t, G_t), \quad (5)$$

where S is the recovery schedule, Q endpoint credits, T IPC/P2P transport generations, L AI leases, E physical fences and worker epochs, C radio commit state, and G multi-home AI ownership. SoftWall builds a private candidate X' and commits it only when

$$\begin{aligned} \text{VALID}(X') \equiv & \text{ExecutableRecovery}(S') \wedge \text{UniqueCredits}(Q', T') \\ & \wedge \text{TimelyAI}(L', E') \wedge \text{CurrentGenerations}(X'). \end{aligned} \quad (6)$$

Otherwise X_t is unchanged. A lease retires only through the matching physical completion fence or a qualified non-launch record. A timeout without evidence quarantines its credit.

SAFETY INVARIANT 1 (CERTIFICATE-TO-EXECUTION REFINEMENT). *For every admitted radio request, all allowed unresolved-outcome branches retain a collision-free conventional path completing by the request expiry. No endpoint, transport, AI, or output credit is reused before matching physical terminal evidence, and each request commits at most one radio result.*

4.4 Decision-window condition

Suppose m_h equal-length recoveries remain compacted at deadline d_h , with guard g_h . At decision time $T_{\text{dec},h}$, let b_{eff} include synchronous control, the AI work-class bound, and completion guard. A sufficient condition for executing AI first is

$$T_{\text{dec},h} + b_{\text{eff}} \leq d_h - g_h - m_h c_h. \quad (7)$$

The right side is the first recovery start in the retained certificate. Equation 7 explains why identical debt and AI class can be safe at 88 ms and unsafe at 89 ms. It is a mode-specific sufficient condition, not a general online scheduling optimum.

PROPOSITION 2 (CURRENT IDLENESS IS INSUFFICIENT). *Consider one non-preemptive recovery lane, a common radio guard boundary D , unresolved recoveries with qualified bounds $c_1, \dots, c_m$, and an external-AI transaction with bound b_{eff} . If mandatory-only execution is feasible but*

$$t + b_{\text{eff}} + \sum_{i=1}^m c_i > D, \quad (8)$$

then a policy that admits AI at time t from current GPU idleness while ignoring unresolved recovery cannot guarantee the radio contract.

PROOF. Choose the allowed realization in which the AI transaction and every recovery take their declared bounds. Because the work is non-preemptive on the contended lane, the last recovery completes after D by Equation 8. The lane may be physically idle at t ; the contradiction comes from future mandatory work rather than queued work. $\square$

This proposition does not say that every sampled debt-blind run must miss. It says that the policy cannot provide the qualified guarantee. A conservative controller can remain safe by validating the initial all-fail admission and physically draining unresolved recovery before AI. SoftWall is needed when the controller wants to place AI first without abandoning that guarantee.

5 Design

5.1 Mandatory-first admission

On radio release, SoftWall reserves conventional recovery before considering an optional endpoint. NeuralRx is admitted only when its whole-path bound fits the fallback cutoff and the endpoint has a generation-matching ring credit. The initial all-fail certificate therefore exists before speculative radio work begins.

5.2 Outcome-driven replanning

All NeuralRx outcomes observed at a common cutoff are applied as one batch. Timely successes delete their debts; failure, lateness, or missing evidence retains them. SoftWall rebuilds the unresolved calendar at the current time and tests visible AI requests against the updated certificate.

Table 2. Feasibility classes.

Class	Meaning
QSU	All-fail radio schedule is executable and at least one qualified AI unit is admissible.
QSN	Radio schedule is executable, but the candidate AI unit cannot safely fit.
MI	The mandatory all-fail schedule itself is infeasible.
UQ	A required bound or lifecycle provenance is absent or failed qualification.

Applying simultaneous successes one by one is unsafe as a decision procedure: an intermediate debt set that never existed can be infeasible at the later host time even when the final observed set is feasible. The batch transition avoids this order dependence.

5.3 AI leasing and launch-time enforcement

AI work is non-preemptive and admitted by work class. A certificate older than the bounded launch control interval is discarded and recomputed. The worker receives an absolute latest-start timestamp and does not launch a CUDA kernel after it. Empty non-launch leases still produce terminal evidence so ownership can retire without guessing.

The transaction charges physical work and the control required to start it. This closes a counterexample in which a fixed [start, end] lease was valid when computed but launch control delayed the kernel into recovery time.

5.4 Distributed ownership and fault containment

One global AI queue can be offered to several RAN homes. A prepare creates at most one unlaunched token per home. Commit grants launch authority only after a current local/global certificate is validated. Reply loss after a broker-side effect makes the request ambiguous; SoftWall does not retry it as a new request. The affected home closes new AI admission while its radio path and other homes continue.

Prepare, abort, commit, and complete carry generations and durable terminal markers in the qualified same-worker reconnect subset. Process replacement and arbitrary crash-window recovery remain outside the paper’s claim.

5.5 Certificate-conformant execution

The executor performs only the next non-preemptive action from the returned certificate, observes the event, and replans. It does not reorder recovery around AI using an unverified host-side plan. CUDA events and worker-side synchronization establish physical completion before IPC buffers, ring credits, leases, or radio output state are reused.

5.6 Feasibility classification

UQ is distinct from overload. A fifth debt can be MI under known bounds, whereas a node that violates the NeuralRx bound is UQ even when arithmetic says the schedule would fit.

6 Implementation

SoftWall runs on NVIDIA A100 nodes with MIG disabled and legacy MPS enabled. The evaluated topology uses local and remote TensorRT NeuralRx endpoints, CUDA IPC and NVLink P2P transports, persistent NVIDIA Aerial/cuPHY conventional workers, and Qwen2.5-1.5B prefill as external AI. Multi-home modes send radio requests to a shared conventional-recovery GPU. IPC

provides process isolation and explicit buffer ownership; P2P moves CUDA-resident data across GPUs without converting recovery semantics into an implicit copy-completion assumption.

The controller maintains recovery calendars, endpoint/ring credits, worker epochs, transport generations, global AI ownership, and single-commit state. A candidate scheduler constructs a schedule; an independent verifier checks every job key, release, service duration, deadline, lane overlap, and AI blackout before the runtime can use it. Exact search is a small-state oracle rather than a claimed optimization contribution.

Instrumentation records host release and commit timestamps, request and payload identities, CUDA completion, IPC/P2P generations, MPS lifecycle, and ownership transitions. Nsight traces validate GPU ordering for selected modes. Source, protocols, seeds, node exclusions, analyzers, failures, and result hashes are preserved in immutable manifests.

7 Evaluation Methodology

We ask seven questions: (1) why MPS alone is insufficient; (2) whether certificate-preserving transactions execute on the actual NeuralRx-cuPHY-Qwen path; (3) whether specified control/data faults are contained; (4) whether the model predicts unopened physical boundaries; (5) whether certified construction scales beyond exact search; (6) whether retiming yields material AI value over a strong safe baseline; and (7) which lifecycle modes are qualified.

Testbed and workload. Authoritative campaigns use two independent nodes, each with four A100 GPUs and NVLink connectivity. MIG is off and MPS is on. Radio execution uses NVIDIA Aerial/cuPHY and TensorRT NeuralRx; external AI is Qwen2.5-1.5B prefill. The workload replay derives arrivals and token values from four non-overlapping windows of the public BurstGPT trace [15]. The evaluated timing contract is the synthetic period/deadline pair P180/D155; it is not labeled as a production HARQ deadline.

Qualification and experimental discipline. Each authoritative experiment freezes source hash, protocol, seeds, node assignment, bounds, exclusion rules, and analyzer before opening outcomes. Development and holdout runs use different nodes and seeds. A model prediction promoted to a physical boundary test is frozen before execution. Failed, negative, and unqualified runs remain in the ledger. We report zero *observed* violations rather than infer zero probability or WCET from finite samples.

Baselines. The main safe comparison is *certificate-preserving recovery-first*: it uses the same max-radio decisions, initial all-fail radio admission, observable state, bounds, arrivals, exact offline weighted selector, single-commit rule, and qualified workers, but physically drains unresolved recovery before external AI. The comparison therefore isolates the value of AI-first atomic retiming; it is not a test of whether recovery debt exists. MPS-only and debt-blind current-idle admission are diagnostic unsafe baselines. The offline oracle observes the realized trace and supplies an upper bound; it is not implementable.

Reproducibility. The anonymous artifact package contains the source snapshot, machine-readable protocols, manifest hashes, analyzers, exact checker, independent certificate verifier, result summaries, and figure builders needed to reproduce every aggregate from preserved raw records. Hardware campaigns require the proprietary Aerial/cuPHY and TensorRT environment; model/checker, trace, and analysis paths run without it. Node-specific names and absolute paths are removed from the review artifact.

Use of generative AI. Generative-AI tools assisted with language editing, document organization, bibliography normalization, and consistency-checking scripts. The authors designed the system

Table 3. End-to-end variable-class qualification.

Metric	Development	Holdout	Combined
Epochs	600	600	1,200
Actual NeuralRx requests	2,400	2,400	4,800
Timely NeuralRx successes	1,750	1,738	3,488
Physical shared recoveries	650	662	1,312
Qwen completions	544	533	1,077
Radio commits	2,400	2,400	4,800
Deadline misses	0	0	0

and model, fixed the experimental protocols, executed the experiments, inspected failures, and verified every reported claim against hashed artifacts. Generative AI produced no experimental measurement or ground-truth label.

8 Evaluation

8.1 MPS is a mechanism, not the contract

Under separate-client overrun, active-thread caps of 100% and 20% produced 2/1,500 and 1/1,500 deadline misses. A later lifecycle comparison observed 0/10,000 misses for a CPU socket sham and 12/10,000 for the GPU/MPS-client arm. These tests do not establish a universal failure probability. They show that cap selection alone cannot be the certificate used by this paper.

8.2 End-to-end variable-class qualification

The authoritative warm P180/D155 campaign used independent development and holdout nodes with the same frozen source and different seeds. It admitted Qwen contexts from 16 to 512 tokens with class bounds {35, 35, 35, 40, 65, 75} ms.

Maximum observed NeuralRx, recovery, and radio-commit paths were 18.123, 13.465, and 118.022 ms. Every executed Qwen unit remained within its frozen class bound. The certificate rejected 55 context-256 and 68 context-512 requests before launch when current debt made them unsafe. Both nodes exercised bounded certificate revalidation; no Qwen kernel began after its latest-start.

8.3 Fault closure

A finite-state audit checked 51 combinations of epoch, generation, fence, and marker state with no invariant violation. The physical A0–A6 campaign covered clean operation, correlated NeuralRx failure, reply delay before and after physical completion, stale and duplicate responses, terminal channel fault, and continuation after fault.

Across qualified nodes it executed 700 epochs, 2,800 actual NeuralRx requests, 942 shared recoveries, and 2,800 radio commits with no observed deadline miss. It included 40 stale/duplicate replay pairs, four terminal channel faults, and 260 post-fault continuation epochs. One other node violated the NeuralRx whole-path bound and was retained as UQ rather than pooled into the passing result.

8.4 Predicting an unopened physical boundary

The envelope was compared with exact feasibility on 16,023 qualified states and retrospectively with 1,200 physical decisions; both had zero mismatch. We then froze seven physical boundary

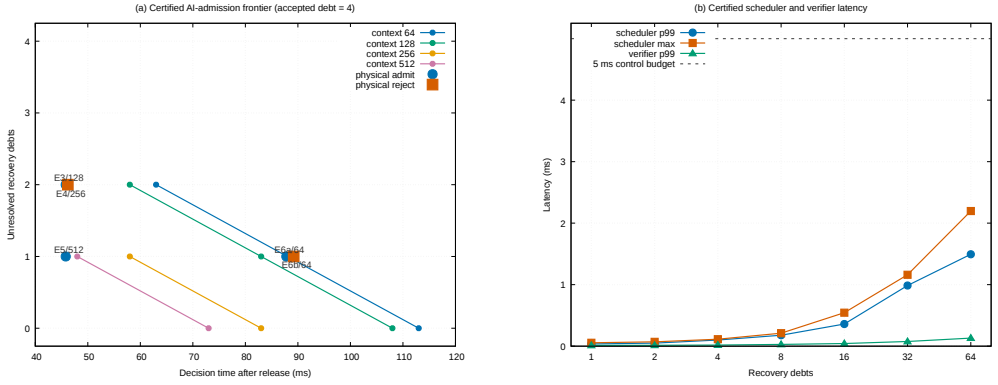


Fig. 2. Qualified AI-admission frontier versus debt, AI class, and decision time (left), and certified-scheduler latency through 64 debts (right). Prespecified physical cases fall on predicted sides of the frontier. Latencies are finite-run measurements, not WCETs.

cases before execution. Every round ran four actual NeuralRx requests and used controlled suffix faults to set unresolved debt, followed by real shared-cuPHY recovery and, when permitted, real Qwen.

Across two independent nodes and 180 rounds, model and runtime agreed in all 180. The run executed 720 NeuralRx requests, 330 recoveries, 90 Qwen units, and 720 commits with no observed deadline miss. At the adjacent decision-time boundary, context-64 at conservative integer time 88 ms was admitted in 30/30 rounds, while 89 ms or later was rejected in 30/30.

8.5 Certified scheduler scalability

The polynomial candidate scheduler tries five deterministic list orders and returns the first schedule accepted by an independent verifier. On 600 heterogeneous small states, exact search accepted 504 and the certified scheduler accepted 494: false-safe was zero and false-conservative was ten. On the qualified mode, decisions matched exact search on all 16,023 states.

For a 2,800-state grid with 1–64 debts and capacities 1/2/4/8, every returned certificate passed the verifier. At 64 debts, candidate decision latency was 1.494 ms p99 and 2.197 ms maximum; verifier p99 was 0.131 ms. These are finite-run CPU measurements.

8.6 What the certificate prevents

Two prespecified C162 cases isolate necessity from throughput. In E4, two recovery debts remain at 45 ms and a context-256 AI request is visible. Mandatory-only recovery is feasible, but admitting the AI transaction yields a bound-respecting completion of 165 ms, 12 ms past the 153 ms radio guard. In E6b, one debt remains and context-64 becomes unsafe at 89 ms: the corresponding completion is 154 ms, one millisecond past the guard. An idle-only debt-blind policy would accept both states; SoftWall classified both as QSN and rejected the AI launch in 30/30 physical rounds per case across two nodes.

These are contract-level counterexamples under Proposition 2. We do not label them observed debt-blind deadline misses because the rejected kernels were not launched. Separate MPS-only diagnostics did observe 2/1,500 and 1/1,500 misses at 100% and 20% active-thread caps, and a lifecycle experiment observed 12/10,000 misses in its GPU/MPS arm versus 0/10,000 in the CPU

Table 4. Safety necessity and AI-first utility are separate comparisons.

Policy	All-fail check	AI before debt clears	Finding
Debt-blind current-idle	No	Yes	Cannot guarantee E4/E6b; unsafe diagnostic
Certificate-preserving recovery-first	Yes	No	Safe conservative comparator; 385,262 timely tokens
SoftWall	Yes	Only with witness	Same 385,262 tokens; predicts safe/unsafe boundary

sham. Together, the evidence distinguishes an actual MPS failure mode from the precise conditional-debt admission that prevents a qualified guarantee.

The counterexamples are relative to the qualified $B_{\text{conv}} = 25$ ms service contract. Holding all other charges fixed, E6b disappears at or below 24 ms; E4 remains until 19 ms, so only E4 survives for $19 < B_{\text{conv}} \leq 24$ ms. The authoritative Q2 recovery-path sample maximum was 13.465 ms (median 3.404 ms), making 25 ms $1.86\times$ that sample maximum rather than $8\times$. A sample maximum is not a replacement service bound: any lower B_{conv} defines an unqualified counterfactual until the complete calendar, workload, placement, and lifecycle mode is requalified. If 13.465 ms were hypothetically promoted to a bound, both selected points would cease to be witnesses. Proposition 2 still applies for every positive recovery charge: a tighter bound moves the false-safe interval to a later decision time rather than making unresolved debt irrelevant, and that new boundary requires prespecified physical validation.

A failure-correlation sweep is not required for this safety conclusion. The contract admits correlated all-fail behavior, so one bound-respecting all-fail branch is sufficient to disprove a guarantee from current idleness. Correlation probabilities matter for expected utility, which we do not claim to improve in the evaluated trace.

8.7 Strong baseline and no material headroom

We compare SoftWall with the certificate-preserving recovery-first system described above. The only semantic difference is whether external AI may execute before unresolved recovery is physically completed. On four non-overlapping BurstGPT calibration windows containing 4,290 requests and 1,129,504 offered tokens, both systems assigned 931 timely requests and 385,262 tokens. Under a sensitivity charging every recovery its full 25 ms bound, SoftWall improved timely token value by only 0.066%, far below the prespecified 5% minimum effect. We therefore did not open a confirmatory throughput holdout.

This negative result narrows the contribution. Conditional capacity exists and changes admission, but the evaluated 1 s AI SLO lets recovery-first service catch up. SoftWall is not presented as a higher-throughput optimizer for this trace.

8.8 Lifecycle qualification

The paper’s lifecycle matrix intentionally contains five qualified or partial subsets and five UQ modes (Table 5). Qualified evidence covers warm persistent operation, a first request after 30 s idle, MPS restart followed by full requalification, mandatory radio continuity during Qwen reload, and same-worker channel reconnect reconciliation. The matrix does not claim availability during restart or reload.

Table 5. Claim-scoped lifecycle envelope. “Partial” means the stated safe subset is qualified while broader availability remains outside the claim.

Mode	Status	Claimed property
Warm persistent	Qualified	Full evaluated contract
First request after 30 s idle	Qualified	Bounded first transaction
MPS restart + requalification	Partial	No reuse before requalification
Qwen reload	Partial	Mandatory radio continuity
Same-worker reconnect	Partial	Generation reconciliation
Process/model cold	UQ	None
5 min idle	UQ	None
30 min idle	UQ	None
GC-on	UQ	None
Worker process replacement	UQ	Development canary only

Process/model cold, 5 min and 30 min idle, GC-on, and worker process replacement remain UQ. A single-node process-replacement development canary passed eight episodes but lacks an independent holdout and is not promoted. Durable journal recovery would be a new mechanism and is outside this paper rather than an unfinished requirement for its bounded claim.

8.9 Production exit gates

A source audit found a stronger intermediate target than P180/D155. Aerial testMAC configures CRC-bearing UL indications at $T_0 + 4.5$ ms; its validator reconstructs slot T_0 and classifies later handler arrival as late. NVIDIA describes testMAC as a developer L2 for a controlled environment, so this is a vendor integration target rather than a field-DU d_MAC.

Under optimistic clean, warm, no-AI conditions, local wait-then-recover and same-GPU speculative paths completed 0/1,000 requests within 4.5 ms. Moving precomputed NeuralRx inputs to GPU1 also completed 0/1,000. Moving channel estimation through CRC behind raw-IQ P2P improved completion to 852/1,000 (3.831 ms median), but retained 148 late samples; disabling Python GC did not remove the tail. Binding cuPHY and caller-owned TensorRT to one stream improved timely completion to 885/1,000 but retained 115 late samples. A diagnostic 300-request stage profile localized that implementation’s tail to cuPHY LS channel estimation: its GPU p50/p99/max were 0.866/4.873/9.592 ms and its correlation with pair wall time was 0.9603. TensorRT host enqueue averaged 7.443 μ s, while the graph itself had a 0.886 ms GPU p99. Forward/backward P2P copies averaged 36.261/17.150 μ s on GPU. Derate-match host-call duration averaged 1,060.105 μ s with a 1,105.380 μ s p99, while its GPU p99 was 0.168 ms. Thus transport and TensorRT were comparatively stable.

We then froze one bounded implementation change: a persistent Fortran-layout complex input removes per-request allocation, and the cuPHY stream orders input assembly before NeuralRx without a host synchronization. A 300-request allocation-only ablation was timely for 299 requests. The prespecified 1,000-request stream-ordered development gate improved to 995 timely completions (3.813 ms median, 3.995 ms p99), with both decoders correct for all 1,000, but five late completions rejected the candidate and did not open the independent holdout. Posthoc attribution found four late units whose conventional completion exceeded 4.5 ms and one whose remote NeuralRx GPU service exceeded it. Pair latency correlated 0.941 with conventional completion and 0.331 with remote NeuralRx service. A profiling-only conventional run identified equalization as its most correlated stage ($r = 0.761$ with conventional total; 1.904 ms maximum), while conventional channel

Table 6. Frozen disjoint-seed Sionna CDL-D/E holdout.

Mode	Conv.	NeuralRx	NeuralRx-only	Conv.-only
CDL-D, 100 ns	153/250	164/250	16	5
CDL-E, 100 ns	155/250	163/250	15	7
Paired aggregate	–	–	31	12

estimation reached 1.295 ms. The remaining gap is therefore a tail across both cuPHY service paths rather than TensorRT, P2P, or remote CE alone. Because 4.5 ms is the mode's D parameter, this implementation fails before scheduling or external AI is considered. Profiling changes timing, so stage values are mechanism evidence rather than qualification or WCET.

The Aerial TDL-A subdomain remains unqualified. Across raw and normalized Aerial TDL-A, public-reference DMRS/MCS/start-symbol settings, one-antenna frequency/time channel execution, and an FP32 engine built as in the public notebook, conventional decoding succeeded in every ten-TB arm while NeuralRx succeeded in none. This localizes the gap to an unsupported model/channel interface or domain rather than a simple power, radio-profile, transmitter, or precision setting.

8.10 Supported-channel NeuralRx result

We next reconstructed the public Sionna 1.0.2/TensorFlow 2.19 contract. Clean MCS7 and default Rayleigh controls passed, and a fixed development matrix identified CDL-D/E at 100 ns as holdout candidates while preserving A/B/C failures. Before opening new payload, slot, and channel seeds, we froze a D/E paired holdout with five Es/No strata and 50 blocks per stratum per model. Both pipelines passed 50/50 at 10 dB. Table 6 reports the full paired outcome.

The aggregate discordant outcomes give a two-sided exact $p = 0.00540$. This is a positive Results claim: optional NeuralRx has conditional radio value in the frozen Sionna CDL-D/E mode. It closes the channel gate only for that finite-sample mode, not TDL-A, field IQ, timing, or production HARQ.

9 Related Work

Table 7 states the distinction at the level of the protected object. The listed systems provide important allocation, fallback, isolation, or SLO mechanisms; the missing state is the unresolved conventional obligation created by an optional receiver for the same TB.

AI-and-RAN resource sharing. YinYangRAN adjusts MPS fractions between PHY and ML and can use CPU fallback [10]. CloudRIC places RAN requests across heterogeneous accelerators using queue and deadline estimates [11]. Interplay and CAORA use learning-based orchestration and MIG allocation [12, 13]; HAF separates slow placement from fast deadline-aware CPU/GPU allocation [6]. A distributed AI-RAN platform addresses placement across RAN sites [1]. These systems establish shared AI/RAN acceleration, dynamic allocation, fallback, and deadline-aware placement as prior art. SoftWall addresses a narrower state: conventional recovery of the same TB remains a future mandatory obligation until optional NeuralRx resolves.

Slack reclamation and bounded background work. Concordia reserves CPU capacity for vRAN and reclaims the remainder [4]. DARIS combines MPS, streams, and stage boundaries to admit lower-priority DNN work [2]. Real-time GPU systems such as REEF, XSched, and ntaskset provide preemption, isolation, or context-scheduling mechanisms [3, 5, 14]. SoftWall uses rather than claims these principles; its certificate also includes radio recovery semantics, endpoint/transport ownership, and physical completion.

Table 7. Closest-work distinction. “Not exposed” describes the public design and is not a claim that the mechanism could never be extended.

System	Primary protected/allocated object	State not exposed to its scheduler
YinYangRAN	MPS fraction and CPU fallback	Multi-cell unresolved same-TB recovery calendar
CloudRIC	Queued request and accelerator deadline	Recovery jobs that have not entered a queue
Concordia	Reserved/reclaimed vRAN CPU capacity	Optional-PHY outcome-conditioned debt
DARIS, REEF, XSched, nvtaskset	GPU stage, kernel, context, or partition	Radio recovery, transport ownership, and single commit as one certificate
ARCHES, OCUDU	AI/conventional PHY selection or timing class	Cross-home all-fail debt plus atomic external-AI lease
SMEC	RAN/edge SLO coordination	Same-PHY-GPU optional receiver recovery
SoftWall	Same-TB recovery obligation	Explicitly modeled, certified, and carried through physical completion

AI/conventional PHY paths. ARCHES selects neural or conventional PHY experts by channel conditions [9]. OCUDU exposes in-DU AI-RAN timing classes, retains a conventional path, and validates result and lifecycle behavior [8]. These works prevent us from claiming novelty for neural/conventional coexistence or fallback. In the public designs reviewed through September 2026, we did not find the combination of a multi-cell all-fail recovery calendar, atomic external-AI leasing, and a provenance-qualified predictive envelope. This is a scoped literature finding, not proof that no unpublished or differently named work exists.

MEC scheduling. SMEC coordinates application SLOs through decoupled RAN and edge schedulers and uses MPS priority at the MEC GPU [16]. Its AI workloads execute on a separate edge server rather than sharing the PHY GPU with optional same-TB recovery. We do not claim novelty for 5G-aware deadline scheduling, only for the conditional recovery contract evaluated here.

10 Limitations

First, P180/D155 is a synthetic harness contract: P is the harness inter-release period, not an NR slot duration, and we do not map D155 to a production HARQ deadline. The C163 production d_MAC trace builder, clock and provenance checks, request-specific exact/certified bridge, and expiry validator are implemented and pass 24 unit tests, but we lack a target DU/FAPI trace. Once such a trace is available, this path can classify request-specific expiries with the same contract semantics; the resulting timing vector and mode must still be physically requalified. Aerial testMAC exposes a $T_0 + 4.5$ ms UL-indication threshold, but it is a controlled developer L2 rather than the target MAC. Every implementation tested retained violations at that threshold; even the best raw-IQ two-GPU development prototype was timely for only 995/1,000 clean requests, and its failed gate did not open a holdout. We therefore make no production HARQ guarantee.

Second, qualified bounds are finite-sample whole-path bounds. Zero observed violations do not prove a WCET or zero failure probability. A different hardware, software, placement, or lifecycle fingerprint must be requalified.

Third, all authoritative hardware results use the A100 family. Four-GPU and multi-home paths exercise NVLink P2P, but cross-family validity is untested.

Fourth, the fault model excludes arbitrary GPU/driver hang and full process-replacement recovery. Five lifecycle modes remain UQ. In C158 attempt 4, one NeuralRx completion reached 350.948 ms in a 400-request run whose NeuralRx p99 was 8.111 ms, a $43.3\times$ p99 excursion; the run consequently had 4/400 deadline misses. Targeted 30 s-idle, quiescent MPS-restart/requalification, Qwen-reload, and same-worker reconnect campaigns did not reproduce that excursion. Its root cause therefore remains unresolved, and none of those lifecycle results qualifies the C158 tail away.

Fifth, PHY generalization is limited. External Aerial TDL-A development screens retained conventional success on every ten-TB arm and NeuralRx success on none after testing normalization, public-reference radio parameters, exact one-antenna transmission, frequency/time channel execution, and the public notebook's FP32 engine precision. Separately, a frozen Sionna CDL-D/E holdout passed its high-SNR pipeline and conditional-value gates over 500 paired blocks, with 31 NeuralRx-only and 12 conventional-only low-SNR outcomes. This qualifies that exact standardized simulated-channel mode only. It does not establish Aerial TDL-A, recorded field-IQ, or over-the-air generalization, and the D/E mode has not traversed production timing and integrated service-vector qualification.

Finally, the strongest throughput comparison found no material advantage. A workload with tighter, independently justified AI deadlines may expose a benefit, but selecting such a deadline after seeing the current result would be post hoc. The demonstrated operational value is advance classification of safe, no-slack, infeasible, and unqualified modes, plus launch-time rejection of certificate-breaking work. This paper therefore makes a substrate and envelope claim.

11 Conclusion

Optional NeuralRx changes shared-GPU scheduling because its unresolved conventional path is future mandatory work. SoftWall makes that work explicit as recovery debt, protects it with an executable all-fail certificate, and admits external AI only through a transaction whose control, transport, and physical completion preserve the certificate. Its feasibility envelope predicts when a qualified mode is safe-and-useful, safe-without-slack, mandatory-infeasible, or unqualified.

The evaluated A100 modes show that this contract can drive actual NeuralRx, shared cuPHY recovery, and Qwen execution across multiple GPUs and fault transitions. The evaluation also states where the claim ends: MPS is not isolation, finite samples are not WCET, production timing is absent, and the strong baseline leaves no material throughput headroom. These boundaries are part of the result.

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